INF10025 Tutorial 05 Answers

Complete Solutions & Explanations for Week 5

Question 1: Access Relationship Diagrams (ARD) vs Entity Relationship Diagrams (ERD)

ARD: Implementation-level diagram representing physical tables in Microsoft Access. Shows foreign keys, junction tables, and surrogate keys (e.g., AutoNumber) created in the actual database.
ERD: Logical conceptual model representing business entities, natural attributes, identifiers, and business relationship cardinalities before physical implementation.

Question 2: Natural Keys vs Surrogate Keys

Natural Key: A real-world attribute that uniquely identifies an entity instance (e.g., StudentID, Tax File Number, TaskName).
Surrogate Key: An artificially generated unique identifier added to a table (usually an integer/AutoNumber) with no real-world business meaning (e.g., TaskID).

Question 3: Surrogate vs Natural Keys Trade-offs

a. Winner? Neither is universally superior; selection depends on data stability and architecture requirements.
b. Conclusion: Natural keys preserve domain meaning and simplify queries in M:1 chains, but can lead to cascading updates if business keys change. Surrogate keys guarantee immutability and compact integer indexes, but require additional JOINs to retrieve business-meaningful data.

Question 4: Business Rules Differences

Different organizations adopt distinct operational strategies, compliance constraints, workflows, and policy rules suited to their specific business models and competitive positioning.

Question 5: ERD Analysis (TASK & EMPLOYEE)

a. Left to Right: ONE Task is assigned to ONE Employee.
b. Right to Left: ONE Employee may be assigned to MANY Tasks.
c. Foreign Key? Yes. When converted to a Relational Schema, a M:1 relationship places the Foreign Key inside the table at the MANY end (TASK relation).

Question 6: Foreign Key Location

The **TASK** relation will contain the Foreign Key (referencing `EMPLOYEE`).

Question 7: Relational Schema (Natural Keys)

EMPLOYEE (EmpName, Gender)
TASK (TaskName, Description, *EmpName*)
Foreign Key (EmpName) References EMPLOYEE(EmpName)

Question 8: Relational Schema (Surrogate Keys)

EMPLOYEE (EmpId, EmpName, Gender)
TASK (TaskId, TaskName, Description, *EmpId*)
Foreign Key (EmpId) References EMPLOYEE(EmpId)

Question 9: SQL CREATE TABLE Statements

```
CREATE TABLE EMPLOYEE (
    EmpId NUMBER PRIMARY KEY,
    EmpName VARCHAR(50) NOT NULL,
    Gender VARCHAR(10)
);

CREATE TABLE TASK (
    TaskId NUMBER PRIMARY KEY,
    TaskName VARCHAR(50) NOT NULL,
    Description VARCHAR(200),
    EmpId NUMBER,
    FOREIGN KEY (EmpId) REFERENCES EMPLOYEE(EmpId)
);
```

Question 10: SQL INSERT Statements

```
INSERT INTO EMPLOYEE VALUES (1, 'Fred Bloggs', 'Male');
INSERT INTO EMPLOYEE VALUES (2, 'Sue Davis', 'Female');
INSERT INTO EMPLOYEE VALUES (3, 'Emma Moore', 'Female');

INSERT INTO TASK VALUES (1, 'Cabling', 'Install Network Cable', 2);
INSERT INTO TASK VALUES (2, 'Build PCs', 'Build 25 PCs for new office', 3);
```

Question 11: SQL Validity Check

a. SELECT Empname, Gender → **Invalid** (Missing `FROM` clause)
b. SELECT Employee → **Invalid** (Table name used in `SELECT` without column or `FROM`)
c. SELECT * FROM Employee → **Valid**
d. SELECT Fred FROM Employee → **Invalid** (Unless 'Fred' is a column name)
e. SELECT ALL FROM Employee → **Invalid** (Incorrect syntax, requires column specification like `SELECT *`)
f. SELECT EmpName FROM Employee WHERE Gender Is M → **Invalid** (Requires quotes `'M'` and `=` operator instead of `Is`)
g. SELECT EmpName, Gender ORDER BY EmpName DESC → **Invalid** (Missing `FROM` clause)
h. SELECT * FROM Employee ORDER BY EmpName → **Valid**
i. SELECT Empid, EmpName, Gender WHERE Gender ='Female' ORDER BY EmpId ASC → **Invalid** (Missing `FROM` clause)

Question 12: CREATE TABLE Behavior

a. 1 table (`Person`) is created.
b. An error occurs indicating that the table name already exists in the database schema.